Overall Learning Progress Report

Week 1 — Python + OOP + Code Quality

Objective: Focused on building strong Python fundamentals, clean coding practices, and structured application development.

Topics Covered

- Python environment setup and virtual environments
- VS Code configuration and command-line basics
- Git and GitHub workflow
- Core Python concepts:
  - Variables and data types
  - Lists, tuples, dictionaries, and sets
  - Conditions and loops
  - Functions and modular programming
  - File handling with text and CSV files
  - Exception handling
- Code quality practices:
  - Type hints
  - Docstrings
  - Ruff and Black tools
  - Data validation using Pydantic
  - Single Responsibility Principle (SRP)
- Object-Oriented Programming:
  - Classes and objects
  - Inheritance and encapsulation
  - Project structuring using modules and packages
  - Introduction to Factory and Singleton design patterns

Project Completed — Inventory Management Package

- Reads inventory data from CSV files
- Validates data using Pydantic
- Logs errors effectively
- Generates low stock inventory reports

Overall Progress

Successfully completed foundational Python and OOP learning with practical implementation through the Inventory Management Package project. Improved coding standards, modular development practices, and project organization skills.

Week 2 — Testing + FastAPI

Objective: Focused on building APIs using FastAPI, database integration, and implementing testing practices for reliable backend systems.

Topics Covered

- REST API and HTTP fundamentals
- FastAPI application setup and routing
- Pydantic request and response models
- Swagger UI usage and API documentation
- Open/Closed Principle (OCP)
- SQLAlchemy models and ORM concepts
- PostgreSQL database integration
- CRUD API development
- Testing concepts and testing pyramid
- Test-Driven Development (TDD)
- Pytest fundamentals
- Fixtures and reusable test setup
- Arrange-Act-Assert testing pattern
- Mocking external dependencies
- Parametrized testing
- Error scenario testing

Project Completed — Inventory FastAPI Service

- CRUD APIs for inventory management
- PostgreSQL database integration
- Structured FastAPI architecture
- Automated test coverage using Pytest

Overall Progress

Successfully built and tested backend APIs using FastAPI and PostgreSQL. Gained hands-on experience in automated testing, API architecture, database handling, and scalable backend service development.

Week 3 — Security & API Design

Objective: Focused on building secure, scalable, and production-ready APIs with authentication, authorization, and advanced architecture principles.

Topics Covered

- JWT authentication
- User registration and login workflows
- Token generation and validation
- Role-Based Access Control (RBAC)
  - Admin
  - Manager
  - Staff

- FastAPI dependencies
- Centralized error handling
- Consistent API response handling
- Database relationships and category modeling
- Product-category linking and relational design
- SOLID principles
- Decorator design pattern implementation

Project Completed — Secure Inventory API

- JWT-secured authentication system
- Role-based authorization
- Category and product relationship handling
- Fully tested API endpoints

Overall Progress

Successfully implemented secure API architecture with authentication, authorization, relational database handling, and scalable design principles. Improved understanding of production-level backend development and software architecture practices.

Week 4 — Web Scraping + Data Processing

Objective: Focused on learning web scraping techniques, data extraction workflows, database integration, and handling structured data pipelines using Scrapy and related tools.

Topics Covered

- Introduction to web scraping concepts and crawling workflows
- Scrapy framework fundamentals:
  - Spiders
  - Selectors
  - Items
  - Pipelines
  - Middlewares
  - Project settings
- Creating and managing Scrapy projects
- Data extraction from websites using XPath and CSS selectors
- Pagination and multi-page scraping techniques
- Data cleaning and validation processes
- Exporting scraped data into:
  - JSON
  - CSV
- PostgreSQL integration using:
  - psycopg2
  - SQLAlchemy
- Insert and update database operations

- Timestamp management using:
  - – created_at
  - – updated_at fields
- Introduction to MCP-based architecture concepts
- Crawl4AI repository analysis and tool exposure workflows
- Basic Selenium integration concepts for dynamic websites

Project Completed — Web Scraping & Data Pipeline System

- Created Scrapy spiders for automated data extraction
- Implemented pagination handling for multi-page crawling
- Processed and validated scraped data efficiently
- Stored extracted data in PostgreSQL databases
- Exported data into structured formats like JSON and CSV
- Implemented insert/update database logic with timestamp tracking
- Improved project structure and reusable scraping workflows

Sessions / Knowledge Sharing

- Session on Claude and AI-assisted development workflows
- Understanding practical usage of Claude features in development tasks

Overall Progress

Successfully gained hands-on experience in web scraping, structured data extraction, database integration, and data pipeline development using Scrapy and PostgreSQL. Improved understanding of scalable scraping architecture, data validation, and backend data processing workflows.

Week 5 — Selenium + Playwright + Advanced Scraping

Objective: Focused on learning browser automation, handling dynamic websites, improving scraping workflows using Selenium and Playwright, and integrating scraped data into structured backend pipelines.

Topics Covered

- Selenium fundamentals:
  - – Browser automation
  - – Element handling
  - – Page navigation
  - – Waits and locators
  - – Screenshots and cookies
  - – Headers and browser configurations
- Handling:
  - – Dynamic JavaScript-rendered content
  - – Infinite scrolling
  - – Iframes
  - – Content-type based PDF downloads

- Playwright fundamentals:
  - Browser contexts
  - Page automation
  - Dynamic selectors
  - Link extraction
  - Image scraping
  - Playwright Codegen usage
- Web scraping optimization:
  - Blocking unnecessary resources
  - Managing request handling
  - Improving scraping efficiency
- Data integration workflows:
  - Integrating scraping scripts with pipelines and models
  - PostgreSQL data storage workflows
- GitHub repository research:
  - MCP-based scraping architecture exploration
  - Crawl4AI and related repository analysis

Project Completed — Selenium & Playwright Scraping Automation System

- Scraped QuickSpecs category data using Selenium
- Automated PDF downloads and local file storage
- Integrated scraping workflows with PostgreSQL pipelines
- Explored Playwright for dynamic website automation
- Implemented browser automation for dynamic content extraction
- Improved scraping efficiency using optimized browser handling techniques

Sessions / Knowledge Sharing

- Conducted R&D; on MCP and Crawl4AI-related repositories
- Explored AI-assisted development workflows and scraping automation strategies

Overall Progress

Successfully gained hands-on experience with Selenium and Playwright for browser automation and dynamic web scraping. Improved understanding of advanced scraping workflows, PDF handling, automation optimization, and integration of scraping pipelines with backend systems and PostgreSQL databases.

Week 6 — Playwright + PDF Data Extraction

Objective: Focused on advanced web scraping using Playwright, structured extraction from AMD and HPE specification sources, PDF parsing workflows, regex-based data extraction, and improving reusable scraping architecture for hardware specification datasets.

Topics Covered

- Advanced Playwright scraping workflows

- Handling:
  - Cookie popups
  - Hidden buttons
  - Dynamic page interactions
  - Pagination handling
- Nested loop extraction techniques for structured specification data
- Scraping static and dynamic product URLs
- SQLAlchemy pipeline integration for scraped data storage
- CSV export workflows
- PDF data extraction using:
  - PyPDF
  - PdfReader
- Regex fundamentals for structured text extraction
- Parsing hardware specification fields from PDFs
- Handling inconsistent PDF structures and layouts
- Generalizing extraction logic across multiple PDF formats

Project Completed — AMD & HPE Data Extraction Pipeline

- Scraped AMD server processor specification data using Playwright
- Extracted fields such as:
  - host_url
  - title
  - image_url
  - clock speeds
  - cache
  - memory specifications
  - detailed processor specifications
- Processed individual product pages using nested loop extraction
- Stored scraped data into:
  - CSV files
  - PostgreSQL databases using SQLAlchemy pipelines
- Extracted structured fields from HPE PDF documents including:
  - pdf_path
  - server_description
  - generation
  - dimm_slots
  - maximum_capacity
  - form_factor
  - chipset
- Implemented regex-based parsing logic for semi-structured PDF content

Overall Progress

Successfully gained practical experience in advanced Playwright scraping, structured hardware specification extraction, PDF parsing, and regex-based data processing. Improved understanding of scalable scraping pipelines, SQLAlchemy-based storage workflows, and handling semi-structured PDF data across varying document formats.

Progress for the Month

- Strong foundation in Python fundamentals and structured databases using PostgreSQL.
- Built web applications and REST APIs using FastAPI.
- Good understanding of unit testing and automated testing using Pytest.
- Strong hands-on experience with Selenium and Playwright for scraping and browser automation.
- Experience in web scraping using Scrapy and scrapy-selenium.
- Worked on structured data extraction, PDF parsing, and scraping automation workflows.
- Kickstarted AI learning alongside scraping development.
- Gained knowledge about MCP concepts and conducted R&D; on MCP integration with Memory.net and Crawl4AI-related workflows.

Skills Progression — Week by Week

Week	Focus Area	Project	Core Skills
Week 1	Python + OOP	Inventory Mgmt Package	OOP, Pydantic, Design Patterns
Week 2	FastAPI + Testing	Inventory FastAPI Service	FastAPI, PostgreSQL, Pytest, TDD
Week 3	Security & Design	Secure Inventory API	JWT, RBAC, SOLID, Decorators
Week 4	Web Scraping	Scraping & Data Pipeline	Scrapy, XPath, SQLAlchemy, MCP
Week 5	Browser Automation	Selenium/Playwright Automation	Selenium, Playwright, PDF Handling
Week 6	PDF Extraction	AMD & HPE Extraction Pipeline	Playwright, PyPDF, Regex